

1. Install an antivirus and perform a full scan

Since you're on Windows 11, you already have Microsoft Defender Antivirus built into Windows, so you don't necessarily need to install a third-party antivirus just to complete the practical.

If your trainer specifically requires a third-party product, use one that is currently offered by its vendor and don't install multiple real-time antivirus products simultaneously.

For the assignment, you can use Avast Free Antivirus/Avast One. Avast currently advertises real-time protection for Windows 11 and says its protection analyzes unknown files before they reach the PC.

Avast official Windows antivirus page

After installation

1. Open the antivirus.
2. Update its virus/security definitions.
3. Select Full Scan.
4. Start the scan.
5. Wait for it to finish.
6. Screenshot the results.

Report result

- Don't invent the numbers. Record exactly what your antivirus reports.

2. Compare three antivirus tools

Here's a report-ready comparison based on current vendor information.

Antivirus	Key features	Real-time protection	Updates
Avast	Malware protection,	Yes	Security updates are

	web/scam protection and real-time file analysis		delivered regularly; Avast advertises real-time protection for Windows.
Kaspersky	Malware protection and security monitoring across supported platforms	Yes, depending on product/platform	Uses regularly updated security databases/components
Quick Heal	Virus/malware protection, ransomware protection and continuous monitoring	Yes	Automatic updates are supported; Quick Heal documentation says updates include new virus signatures and recommends keeping the product updated.

3. Test the antivirus with EICAR

This is one of the best parts of the practical because you don't need a real virus.

The EICAR Anti-Virus Test File was specifically created to test antivirus software without using real malware. EICAR states that it is not a virus, but compatible antivirus products should detect it as a test threat.

Use the official EICAR page

EICAR official Anti-Malware Test File download page

On that page you'll find options including:

- eicar.com
- eicar.com.txt
- eicar.com.zip

What to do

1. Make sure your antivirus is running.
2. Open the official EICAR test-file page.
3. Attempt to download the test file.
4. Observe what happens.

Your antivirus may:

- block the download,
- display a malware/test alert,
- quarantine the file,
- delete/remove it,
- or allow it temporarily and detect it during scanning.

4. How real-time protection works

Real-time protection means the antivirus continuously monitors activity instead of waiting for you to manually start a scan.

A simplified process is:

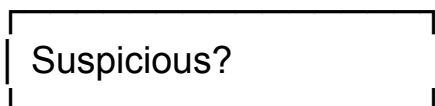
You download/open a file



Antivirus intercepts/checks activity



Signature/reputation/behavior analysis



Yes

No

Block/Quarantine

Allow

Scenario

You download:

- game.apk

The security software can examine the file and its characteristics before or during execution. If it identifies the application as malicious or suspicious, it can warn you or prevent the activity.

This is different from a manual full scan, where you deliberately tell the antivirus to examine files already stored on the device.

Quick Heal, for example, documents continuous virus monitoring for sources including Internet downloads and file execution. Avast likewise describes real-time analysis of unknown files on Windows.

Important distinction

Real-time protection isn't a guarantee that every malicious file will be detected. Modern security products combine multiple techniques because malware can be new, modified, obfuscated, or otherwise difficult to identify.

5. Five malware-safety best practices

1. Keep the operating system updated

- Install Windows, Android, browser, and application security updates.
- Updates often fix vulnerabilities that attackers could otherwise exploit.

2. Keep antivirus protection enabled

- Keep real-time protection turned on and allow security definitions/components to update automatically.
- An antivirus with outdated definitions has less information available for identifying known threats.

3. Download software from trustworthy sources

- Avoid random websites offering "cracked," "patched," or modified applications.
- For Android, prefer Google Play and the developer's legitimate distribution channels rather than unknown APK sites.

4. Be careful with email attachments and links

- Don't automatically open an attachment simply because it arrived in an email.
- Check the sender, context, file type, and destination of links before opening them.

5. Maintain backups

- Keep important documents backed up separately from your main computer.
- A backup can be particularly valuable if ransomware or another destructive malware incident affects the original files.